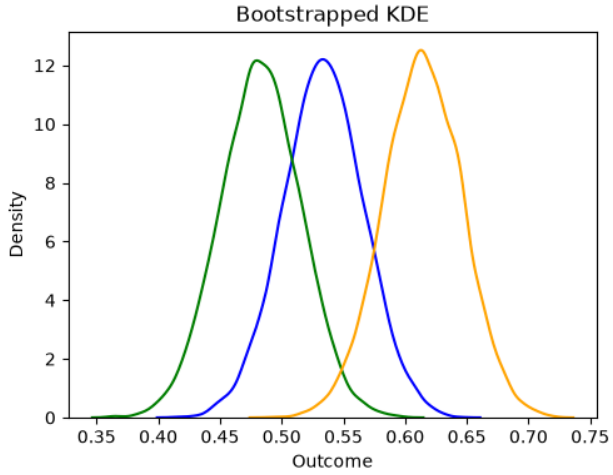
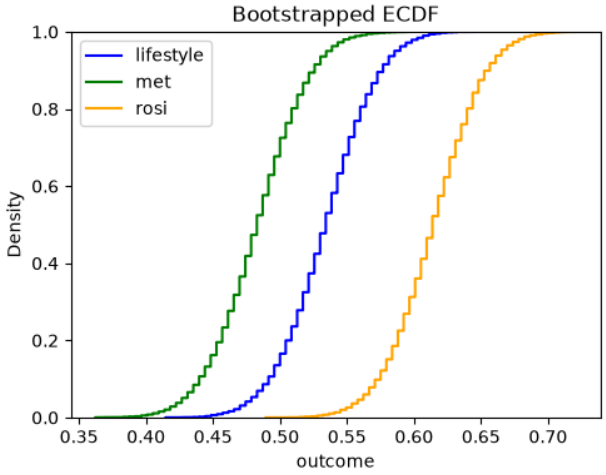
After examining the trial, there is statistically significant evidence that rosiglitzalone is an effective treatment in comparison to metformin. First, it was noticed that the rosi treatment did have a higher rate of success compared to the other two treatments. Rosi treatment had an overall success rate of 61.4%. Lifestyle had a success rate of 53.4% and met had a success rate of 48.3%. These results are pictured in the table below. In order to verify these results, a bootstrapping simulation was performed with 10,000 replicates. The resampling was done separately within each arm, which preserved randomness.

treatment	success rate
lifestyle	0.534
met	0.483
rosi	0.614

During the bootstrapping process, each resample had the same number of participants as the original group. Three pairwise differences in success rate were calculated: met vs. rosi, rosi vs. lifestyle, and met vs. lifestyle. The bootstrapped distributions reinforced the pattern from the original table: rosi had the rightmost KDE density curve, and had an ECDF curve located to the right of the met and lifestyle curves.





The most significant results were the bootstrapped confidence intervals, which are pictured below. The statistically significant intervals are highlighted in green. There was no statistically significant evidence that metformin is more or less effective than lifestyle changes. As seen below, rosi was a more effective treatment when compared to met at all 3 intervals. Because the differences are calculated as the second treatment subtracted from the first

treatment, the fully negative intervals in that row indicate this. There was also a statistically significant difference between rosi and lifestyle at the 90% interval. However, because this pattern only occurred at the 90% interval, it is difficult to conclude that there was a significant difference between rosi and lifestyle. Overall, the study shows that rosiglitzalone was significantly more effective at treating diabetes when compared to metformin, and that there was not enough evidence to support significant differences in success between metformin and lifestyle and rosi and lifestyle.

	90% CI	95% CI	99% CI
met vs. rosi	[-0.208 -0.058]	[-0.221 -0.041]	[-0.247 -0.015]
met vs. lifestyle	[-0.129 0.026]	[-0.141 0.039]	[-0.172 0.065]
rosi vs. lifestyle	[0.003 0.157]	[-0.01 0.17]	[-0.04 0.195]